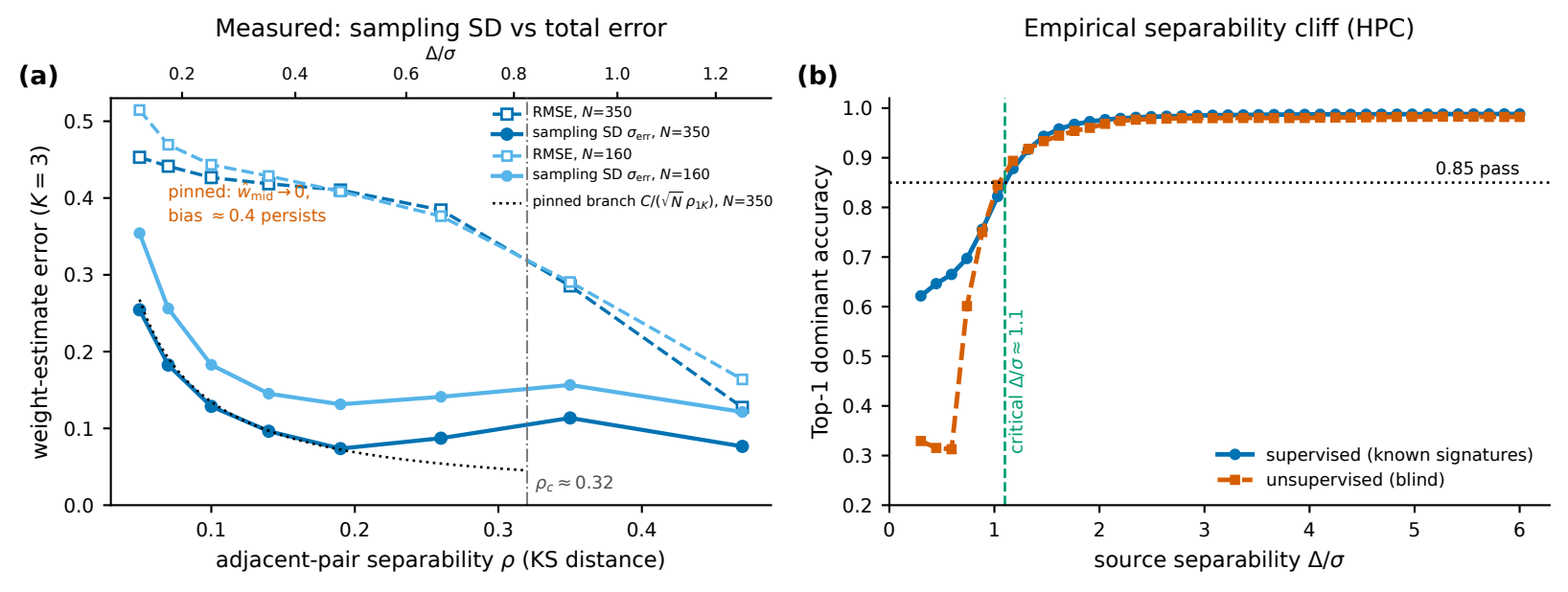
Measured: sampling SD vs total error
(a)
Δ/σ
0.2
0.4
0.6
0.8
1.0
1.2
RMSE, N=350
sampling SD σ_err, N=350
RMSE, N=160
sampling SD σ_err, N=160
pinned branch C/(√N ρ_1K), N=350
pinned: w_mid → 0,
bias ≈ 0.4 persists
ρ_c ≈ 0.32
weight-estimate error (K = 3)
adjacent-pair separability ρ (KS distance)
Empirical separability cliff (HPC)
(b)
0.85 pass
critical Δ/σ ≈ 1.1
supervised (known signatures)
unsupervised (blind)
Top-1 dominant accuracy
source separability Δ/σ